GE Medical Systems

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Technical Publication

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Revision 3

GE Medical Systems
Proprietary ETC to S/A Slipring Conversion

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- **FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.**
- **VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH NICHT ZU RATE GEZOGEN UND VERSTANDEN WURDE.**
- **WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.**

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- **ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS.**
- **SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.**
- **NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.**
- **LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.**

ATENÇÃO

- **ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.**
- **SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.**
- **NÃO TENTE REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.**
- **O NÃO CUMPRIMENTO DESTE AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A‘ CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.**

AVVERTENZA

- **IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.**
- **SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.**
- **SI PROCEDA ALLA MANUTENZIONE DELL’APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.**
- **NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL’ADDETTO ALLA MANUTENZIONE, ALL’UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.**

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14 July 1993

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It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

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End of Section

Revision History

Revision	Date	Reason for change
0	1/31/01	Document Release
1	3/20/01	Added Troubleshooting and Early HSA Gantry Information
2	7/31/01	Improvements added to procedural graphics
3	6/25/02	Confusion of Field Personnel related to ground strap.

List of Effected Pages

PAGES	REVISION	PAGES	REVISION
1 through 54	2		
31	3		

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Table of Figures

Chapter 1
Pre-Installation

Before you get started make sure you check the following;

- Correct conversion kit received.
  - The EMC kit does not contain all parts needed for a Non-EMC gantry.
  - Take inventory of all parts received.
- Proper tools including Torque wrench kit.
- Remove new slip-ring from shipping crate.
  - The new slip-ring needs to adjust to room temperature otherwise it will not properly seat on the bearing flange. This can take about 2 to 4 hours, depending on your geographic location, shipping method and time of year.
  - Do not use the mounting bolts to seat the ring. You will crack the slip-ring and need to order another one.

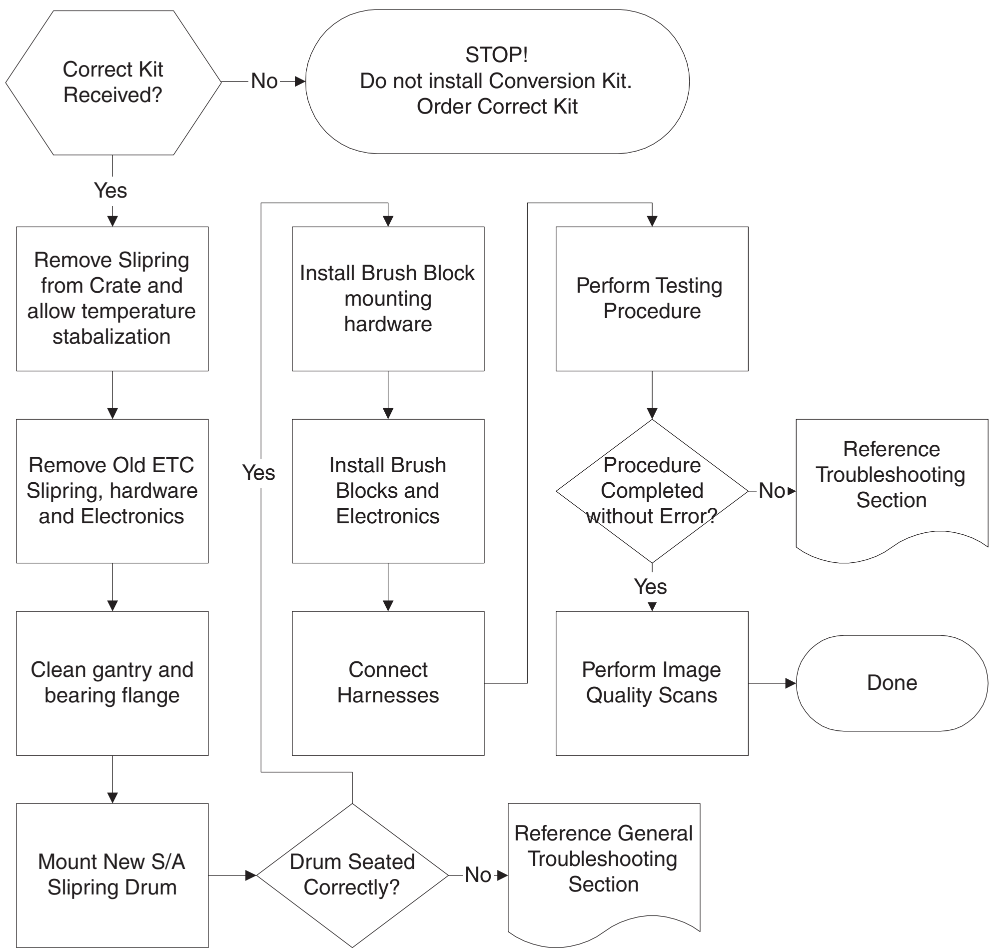


Figure 1-1 Procedure Flowchart

Chapter 2
Replacement Procedures

Section 1.0
Remove Old ETC Hardware

1.1 Required Tools

- 3/32, 1/8, 9/64, 5/32, 3/16, hex key
- 9/64 hex key socket
- Phillips Screwdriver #2
- Flat-blade screwdriver Large and small tip
- 7/16 hex socket
- Torque Wrench Kit
- Hammer and 12 inch long 2 X 4 lumber
- Vacuum cleaner
- 600 Grit sandpaper

1.2 Procedure Details

This procedure requires 2 people for 6 hours and a third person for about 1 hour. The third person will assist in removing and installing the rear gantry cover.

1.2.1 Covers

1.) Position table to lowest elevation.
2.) Turn off facility power to PDU.

DANGER **USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.**

3.) Remove, and set aside, both gantry side covers.
4.) Turn off all three (3) switches on the status control box, on right side of Gantry.
5.) Open top cover, and engage prop rod.
6.) Remove, and set aside, Gantry scan window.
7.) Open Rear cover.
   a.) Locate left hinge arm inside the rear cover, and disconnect the electrical connections leading to the top and rear covers. Reference Figure 2-1 on page 20.
   b.) Disconnect the two (2) wires leading to the rear cover lamp.
   c.) Disconnect the harness leading to the rear cover microphone.
8.) Remove the top cover: (Requires 2 people)
   a.) Engage the prop rod.
   b.) Remove the retaining clip from the end of the top cover gas spring.

c.) Grasp the gas spring, and firmly pull outward.
d.) Let the gas spring swing down and hang from the rear cover support.
e.) Attach the retaining clip to the gas spring for safe keeping.
f.) Remove the rear cover portion of the middle hinge that holds the top cover in place.
g.) Move the top cover to the right, remove it from the rear cover, and set aside.

9.) Remove the Rear Cover: (Requires 3 people)
a.) Remove “C” clips. Reference Figure 2-1.
b.) Grasp the bottom of the cover on each side below the swing doors.
c.) Lift on the Rear Cover Assembly, while the third person guides the pivot pins out of the pivot arm on each side of the cover.

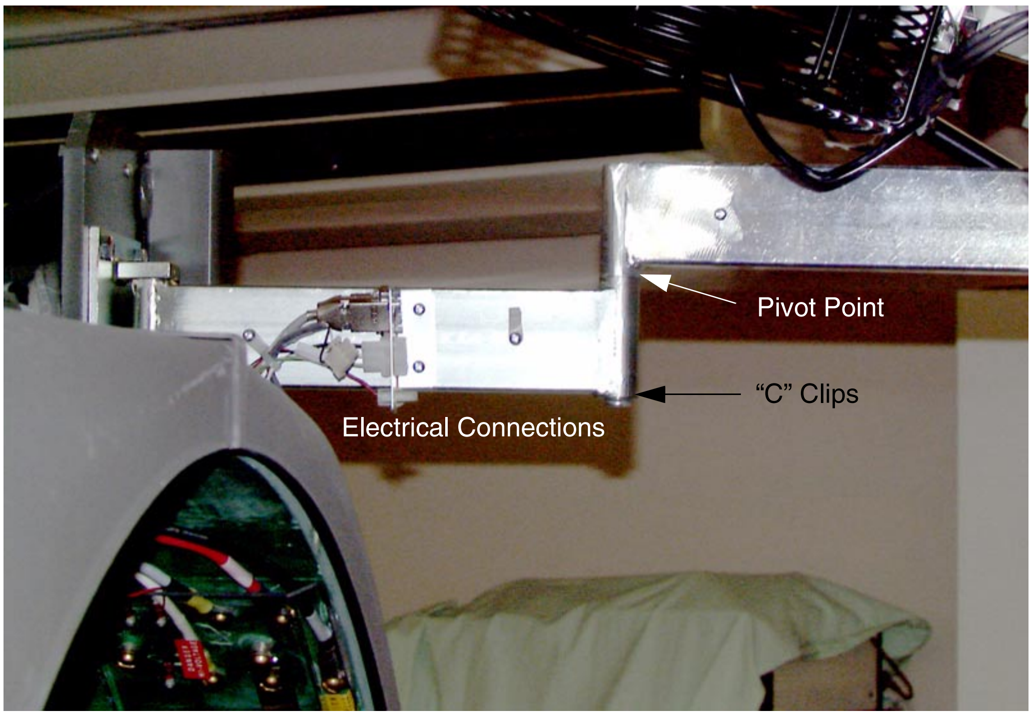


Figure 2-1 Rear Cover Hinge and Connections

d.) Place the Rear Cover on top of a blanket to prevent scratching.
e.) Lean the Cover against the wall, or if space permits, lay the cover horizontally on a blanket on the floor.

10.) Remove the two (2) slipring Covers.

1.2.2 Slipring Buffer Boards

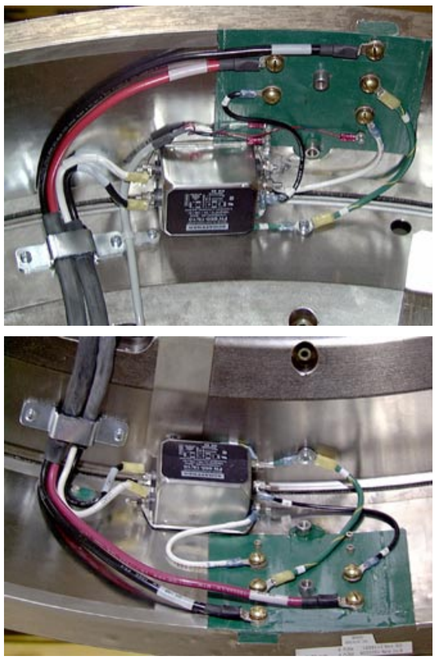
NOTICE **Use proper ESD procedures when handling any circuit boards.**

Comment: *SAVE ALL MOUNTING SCREWS AND WASHERS FOR LATER USE. THIS INCLUDES THE BUFFER BOARD HARDWARE AND THE ROTATING POWER DISTRIBUTION HARDWARE.*

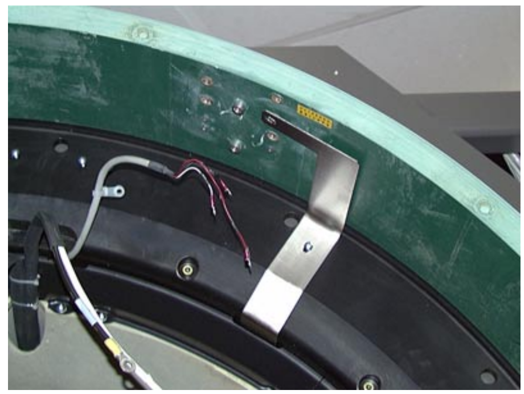
1.) Remove the two BNC Connectors, J1 and J2, from Rotating Buffer Board.
2.) Remove the Rotating Buffer Board and the Rotating Terminator Board:
a.) Remove five (5) 4-40 socket head cap screws and one (1) 1/4 -20 screw from each board.
Remove five (7) 4-40 socket head cap screws and one (1) 1/4 -20 screw from each board.(46-321055P1 Rotating Terminator, 46-3211058G1 Rotating Buffer).

   b.) Remove the boards from the slipring, and place them in anti-static bags.

3.) Remove two (2) Lexan terminal covers for the electrical junctions between the gantry and the slipring.
   a.) Remove the two (2) 1/4 -20 screws that fasten each cover in place.
   b.) Remove both covers, and set aside, for reuse.
4.) For EMC >> Disconnect all cable connections on the inside of the slipring, including the two (2) Ground Strap connections.
   a.) Remove and save the AC Power filter for later use.
   b.) Write down the AC Power Filter orientation and wiring connections.
5.) For Non-EMC >> Disconnect all cable connections on the inside of the slipring, including the four (4) Ground Strap connections on the ring itself.



EMC Slip-ring



Non-EMC Slip-ring

Remove the AC Power Filter and save for later installation.

Figure 2-2 ETC Rotating Power Connections

6.) Remove the Stationary Terminator Board: Reference Figure 2-5 on page 23.
   a.) Remove the two (2) 4-40 socket head cap screws.
   Remove the seven (7) 4-40 socket head cap screws. (46-321052G1 Stationary Terminator).
   b.) Remove the board and place it in an anti-static bag.
7.) Remove the Stationary Buffer Board: Reference Figure 2-5 on page 23.
   a.) Disconnect two (2) BNC connectors, J1,J2, and power connector J3.
   b.) Remove the five (5) 4-40 socket head cap screws.
   Remove the eight (8) 4-40 socket head cap screws. (46-321056G1 Stationary Buffer)
   c.) Remove the board and place it in an anti-static bag.
8.) Remove, and set aside, the one (1) power and two (2) signal Brush Blocks. Reference Figure 2-5 on page 23.
   a.) Disconnect the Power Brush electrical connections.
   b.) Disconnect the grounding strap.
   c.) Four (4) bolts fasten each Brush Block in place.

1.2.3 ETC RF Slip Ring Shoe (Lightspeed QXI Only)

1.) Disconnect the coax cable from the RF shoe. It is a push-on / pull-off type connector. Be careful not to pull the cable out of the connector housing.
2.) Remove the radial adjustment screw for the RF shoe.

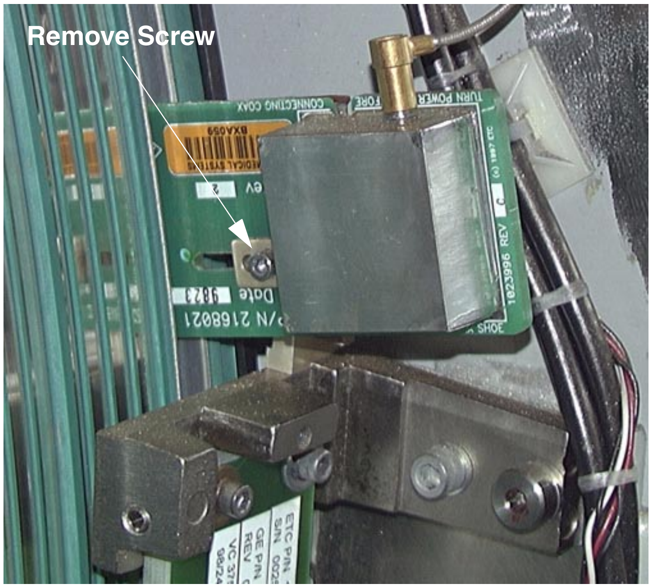


Figure 2-3 ETC RF Shoe

1.2.4 ETC RF Slip Ring Receiver (Lightspeed QXI Only)

1.) Locate the Stationary RF Power Pan on the rear gantry frame, upper Right corner.
2.) Remove the ETC RF Receiver amplifier.

Figure 2-4 ETC Non EMC Stationary Power Pan

1.2.5 ETC RF Slip Ring Tx Balun (Lightspeed QXI Only)

1.) Locate the RF Tx Balun located on the inside of the slip ring, next to the RF Transmitter.
2.) Remove the coaxial RF connector. Be careful when pulling on the snap-on RF coaxial connector.

1.2.6 Mounting Brackets

1.) Remove six (6) Brush Block Brackets: Reference Figure 2-5.
   a.) Remove the shoulder screw and socket head cap screw from each bracket.
   b.) Remove each bracket.

Figure 2-5 Removing ETC Power Block Hardware, Stationary Blocks are Similar.

2.) Open the front cover and remove Home Flag Sensor Cover.

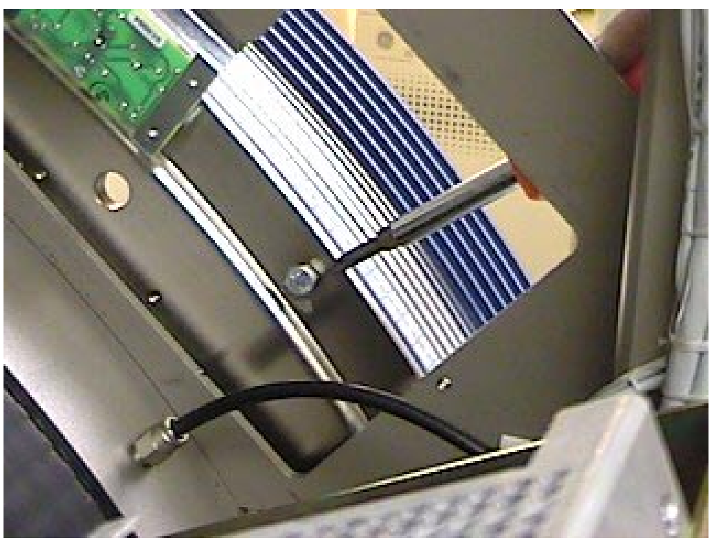


Figure 2-6 Slipring Mounting Bolt Access Port

Note: A raised pilot on the bearing flange helps hold the slipring in place, as long as you apply pressure against the back face of the slipring.

3.) Remove eight (8) bolts from the slipring: (Requires 2 people)
   a.) Rotate the gantry to position one (1) bolt in the access port from step 2 above.
   b.) Remove the bolts from the gantry access port, and set aside for reuse.
   c.) Second person: Push against the back face of the slipring to hold it in place while your partner removes the bolts.

d.) Rotate the Gantry, by hand, until another bolt appears in the access port.

e.) Repeat the procedure to remove a total of 8 bolts.

4.) Remove the slipring assembly:

a.) BOTH people stand on either side of the slipring assembly.

b.) Grasp the slipring with both hands.

c.) Pull the slipring off the bearing flange.

CAUTION **It might be necessary to use a thin flat blade screwdriver as a wedge to separate the slipring from the bearing flange. BE READY TO CATCH THE SLIPRING. THE SLIPRING WEIGHS ABOUT 75 POUNDS.**

NOTICE Potential for slip-ring damage **The epoxy slip-ring is fragile, so handle it with extreme care. DO NOT STAND THE SLIPRING ON END; LAY IT FLAT.**

5.) Carefully remove the dry loctite from the bearing flange surface. This is to ensure a good flat seat for the new slip-ring assembly. Reference Figure 2-7.

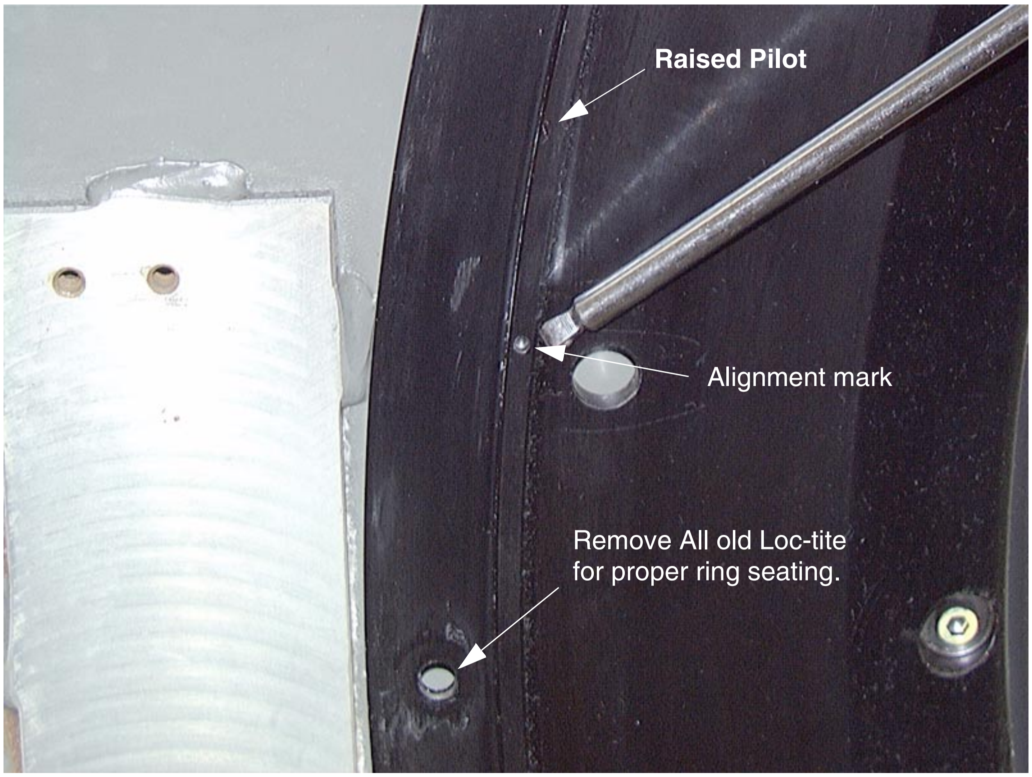


Figure 2-7 Bearing Flange Preparation

Section 2.0
Install New Hardware

2.1 S/A HSDCD Slipring Assembly

Note: The slip-ring is encased in plastic and the plastic should be removed from the shipping crate. This will allow the slip-ring to stabilize at room temperature. The plastic protects the ring from contamination. Slip-ring replacement requires two people, plus a third person for approximately one hour.

2.1.1 Gantry Cleaning

At this point use the HEPA Vacuum cleaner and remove all existing slipring dust from the newly accessible areas of both the stationary and rotating members. Make sure that ALL old loctite has been removed from the bearing flange. Remove any metal burrs to ensure a flat surface for the new slipring.

2.1.2 Installation

1.) Find and remove the hardware and electronics from the shipping crate. (Brush blocks, mounting brackets, Transceiver boards, terminators).
2.) Carefully remove the S/A slipring from the shipping crate.
3.) Remove the plastic bag and inspect the ring for defects. Remove any foreign objects specifically from the rear or mating surface. This is to ensure the female threaded inserts will not be damaged when installing the slipring.

Figure 2-8 Slipring Shipping Crate

NOTICE **Potential for slipring damage. The epoxy slipring is fragile, so handle it with extreme care. DO NOT STAND THE SLIPRING ON END; LAY IT FLAT.**

CAUTION **MAKE SURE THE EMI SHIELD BRAID IS FOLDED AWAY FROM THE SLIPRING BASE. DO NOT PINCH THE EMI SHIELD BRAID BETWEEN THE SLIPRING AND THE BEARING FLANGE. THE SLIPRING WILL NOT SEAT PROPERLY IF THIS OCCURS.**

4.) Replace the slip-ring:

a.) Carefully fold the braided EMI gasket toward the center of the slip-ring drum. This is necessary to prevent the braid from being pinched between the bearing flange and slip-ring drum. A single strand will prevent the slip-ring from properly seating.

b.) With clean cotton gloves hold the slipring next to bearing flange.

CAUTION Possible Ring Contamination

Do not use latex gloves. The powder from the gloves can contaminate the slipring.

c.) Rotate the bearing, and/or slip-ring, to align the black line on the inside of slip-ring with the small hole on the bearing flange pilot. Reference Figure 2-9.

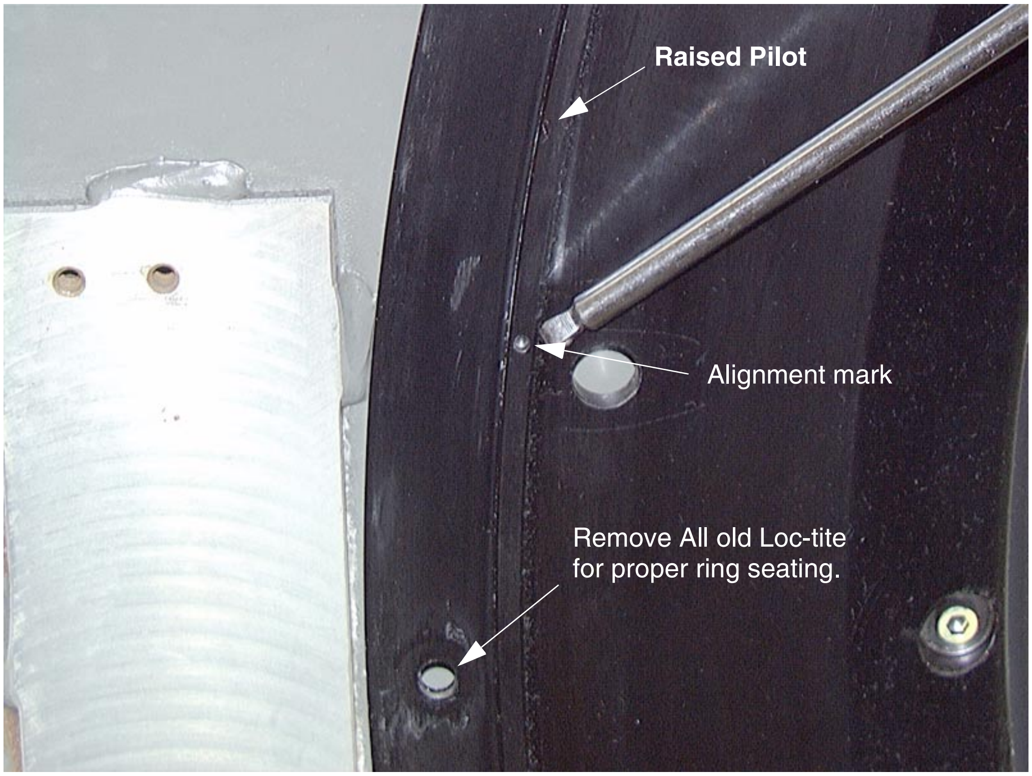


Figure 2-9 Bearing Flange

5.) Fasten slip-ring to Bearing Flange: (Requires 2 people) Reference Figure 2-10.

a.) Second person: Press the slipring flush against the Bearing Flange while your partner replaces the bolts.

b.) Install four (4) bolts in every other hole first.

c.) Snug these bolts to 3 ft.-lbs.

NOTICE

DO NOT USE THE MOUNTING BOLTS TO SEAT THE RING. DOING SO WILL DAMAGE THE INSERTS, REQUIRING ANOTHER RING REPLACEMENT.

d.) Very carefully use the 2 x 4 lumber wrapped in a towel to aid in seating the ring on the bearing flange. Make sure the block is flat against the ring.

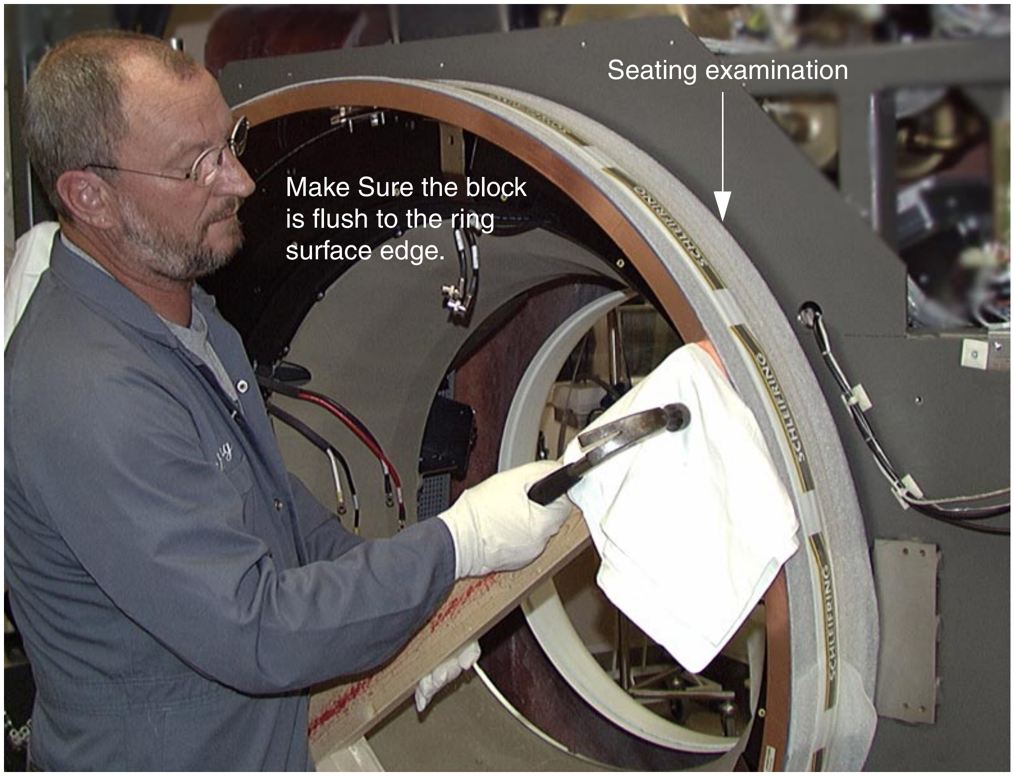


Figure 2-10 Slip-ring Seating

e.) Alternate method is to use clamps and a rubber mallet. Make sure there is padding or wood shims between the clamp and ring surfaces to prevent damage to the ring. Reference Figure 2-11.

Using this method, you need to move the clamp at 30 degree increments until ring is fully seated on the bearing flange.

f.) Using either method, snug the mounting bolts as the ring is drawn onto the bearing flange.

Note: This is a press fit assembly. You need to maintain equal spacing around the entire circumference to correctly seat the slipring.

CAUTION **MAKE SURE THE EMI SHIELD BRAID IS FOLDED AWAY FROM THE SLIPRING BASE. DO NOT PINCH THE EMI SHIELD BRAID BETWEEN THE SLIPRING AND THE BEARING FLANGE. THE SLIPRING WILL NOT SEAT PROPERLY IF THIS OCCURS.**

g.) Examine the clearance between the bearing flange and ring.

h.) Snug the four (4) bolts to retain the ring seating.

Figure 2-11 Slip-ring Seating Alternate Method

CAUTION **MAKE SURE THE EMI SHIELD BRAID IS FOLDED AWAY FROM THE SLIPRING BASE. DO NOT PINCH THE EMI SHIELD BRAID BETWEEN THE SLIPRING AND THE BEARING FLANGE. THE SLIPRING WILL NOT SEAT PROPERLY IF THIS OCCURS.**

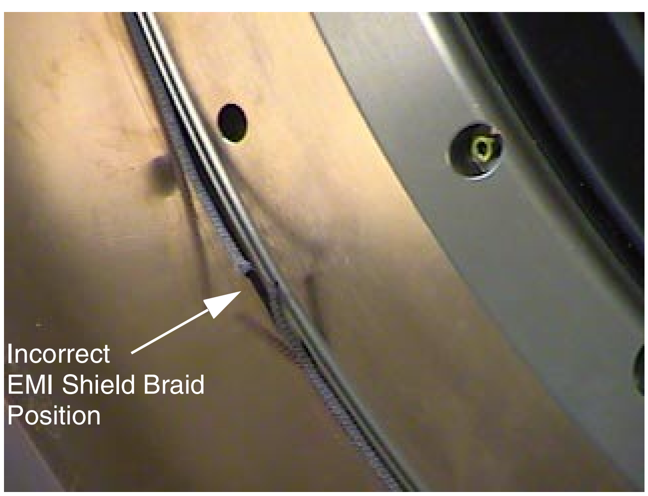


Figure 2-12 EMI Shield Braid

i.) Apply a small amount of 242-Loctite to each of four (4) remaining slipring bolts.

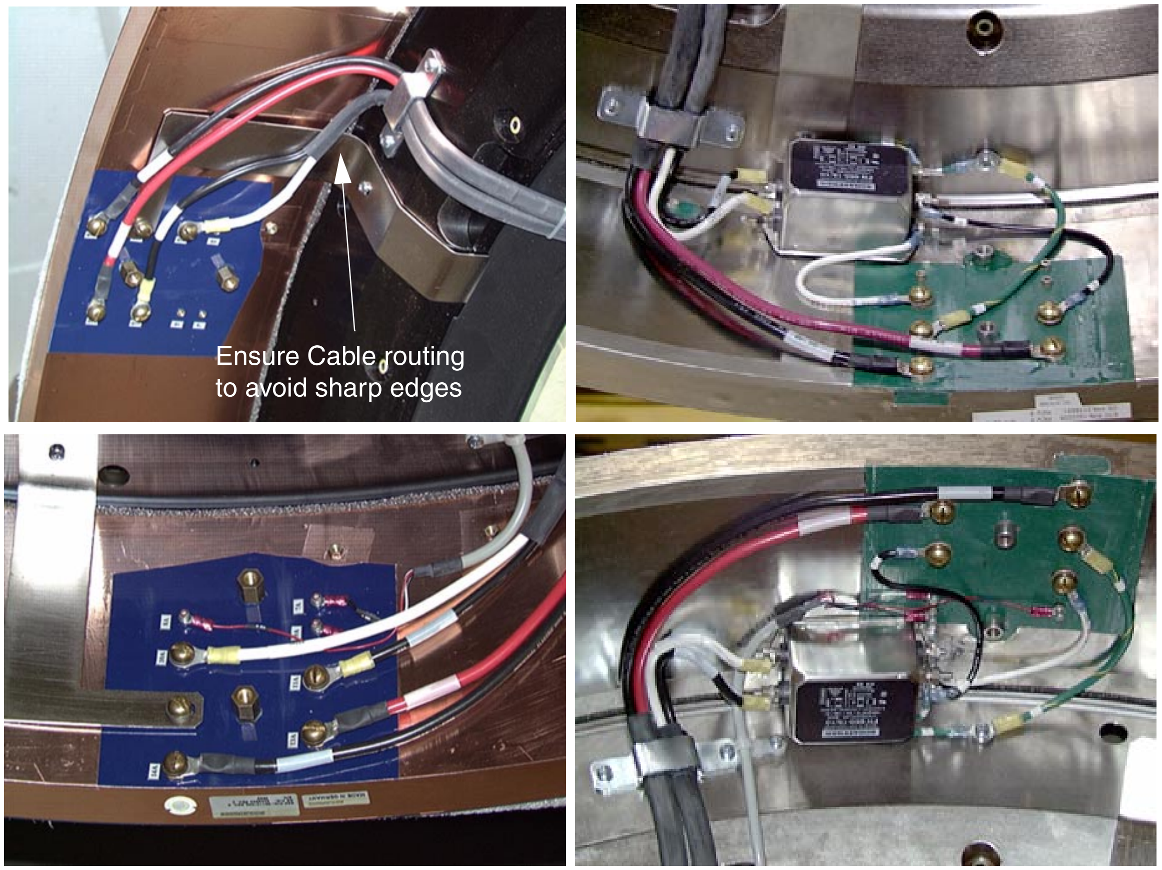
j.) Install the second set of four (4) bolts, and tighten to 5 ft.-lbs.

k.) Remove the first set of four (4) bolts and apply 242-Loctite.

l.) Now install the remaining four (4) bolts and tighten to 5 ft.-lbs.

6.) Recheck the torque of all eight (8) bolts: 5 ft.-lbs. (0.0048 m-kg).

7.) Connect all power and signal wiring to the inside diameter of the slip-ring. Use the hardware from Comment:, on page 20 .



Non-EMC Slip-ring EMC Slip-ring

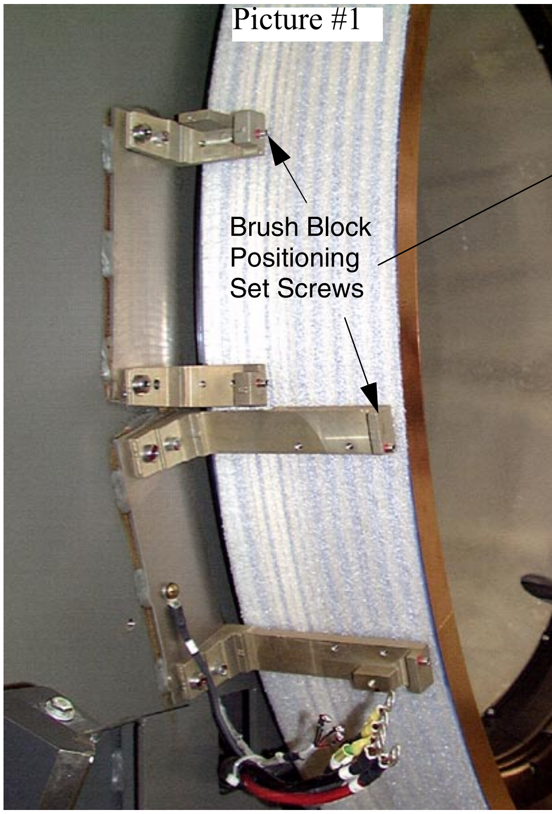
Figure 2-13 Rotating Power Harness Connections

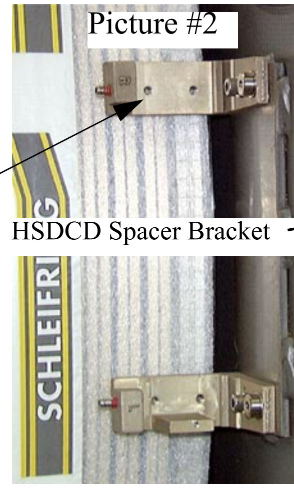
NOTICE **Make sure all connections are tight and safety covers installed.**

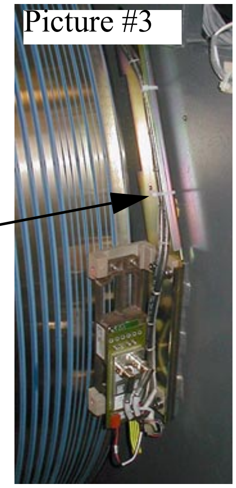
For very early HSA Gantries, re-terminate the 3 low power leads, rings 7, 8, 9. Use terminals supplied in auxiliary parts kit.

8.) Install the Brush Block mounting brackets and HSDCD bracket. Reference Figure 2-14.

Install the HSDCD bracket used on LightSpeed systems. This HSDCD bracket is also used on EMC and Non-EMC gantry's as a spacer for proper Transceiver circuit board alignment. Reference Figure 2-15 for location.







Use new Brackets. Each bracket has an “L” or an “R” stamped on it. The “L” is the top brackets in picture #1 and the bottom bracket in picture #2. Use Loc-tite 242 on bracket mounting screws. Make sure you install the HSDCD bracket on ALL Gantries. It is a spacer in EMC and Non-EMC Kits. HSDCD spacer is installed under the brush block brackets. Picture #2 does not show the spacer. Picture #3 shows the final assembly.

Figure 2-14 Brush Block Bracket Installation

9.) Install HSDCD Gap Tool storage bracket. Reference Figure 2-15.

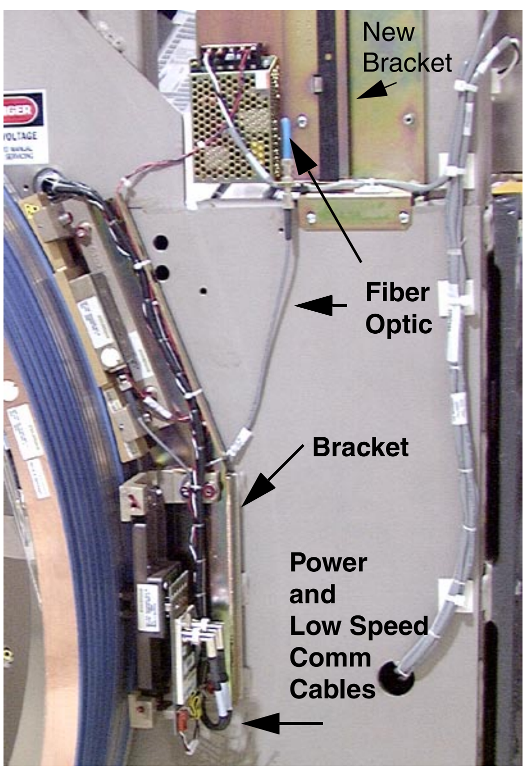


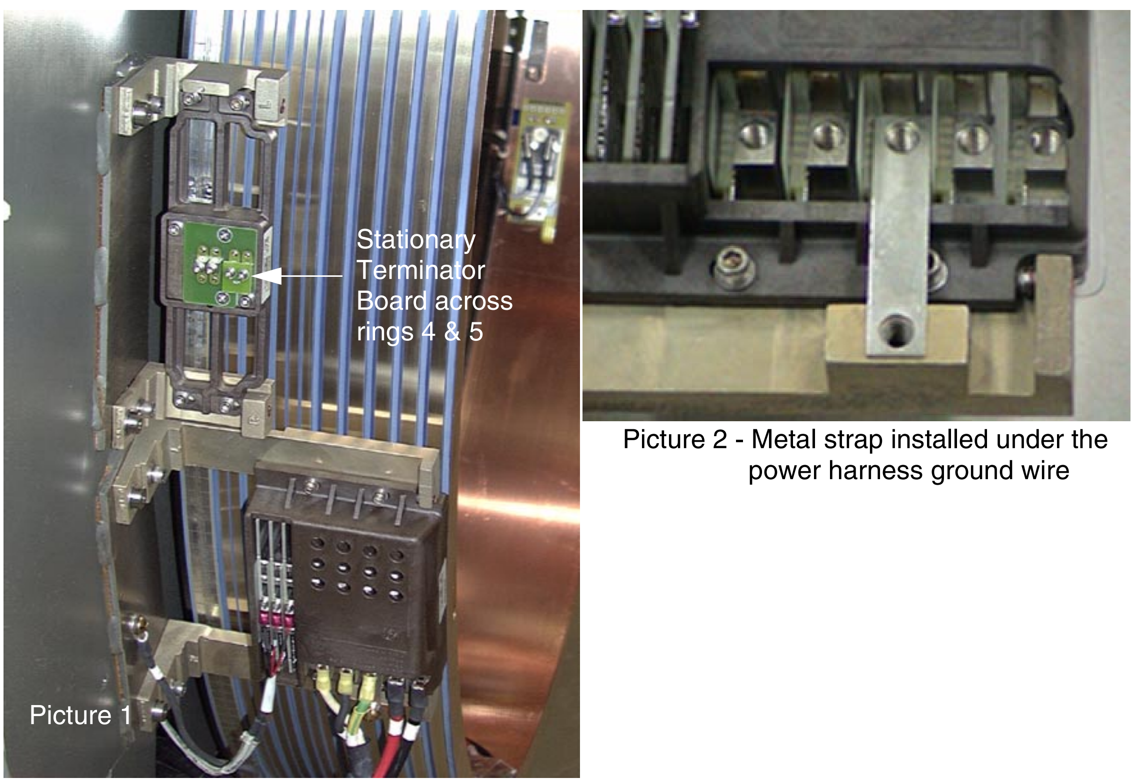
Figure 2-15 Lightspeed QXI Stationary Bracket and Wire Routing

2.2 S/A HSDCD Power Brush Block Assembly

CAUTION **DO NOT APPLY SIDEWAYS FORCE TO THE BRUSH BLOCK ASSEMBLIES. THE BRUSHES ARE BRITTLE AND WILL BREAK.**

1.) The Gantry contains three (3) brush blocks: two (2) for signal and one (1) for power.
2.) Install Power Brush Block:
   a.) Hold the Brush Block in position, and align the brush tips with the slipring tracks.
   b.) Compress the brush tip springs against the slipring tracks, and hold the brush block against the brackets.
   c.) Install all four (4) screws, but do not tighten.
   d.) Adjust the brush blocks to center the top and bottom of the brush tips on the ring.
   e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)
   f.) Adjust the positioning set screws on the bracket to contact the brush block assembly. Set screws contain pre-loaded Loctite. Reference Figure 2-14 on page 30.
3.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.
4.) Connect the power wiring. Make sure you install the grounding strap.
   - The metal strap shown in Figure 2-16 picture 2 is installed under the power harness ground wire.
   - Failure to install this ground strap will result in intermittent communication and other errors.

CAUTION **For very early HSA Gantries, re-terminate the 3 low power leads, rings 7, 8, 9. Use terminals supplied in auxiliary parts kit.**



Picture 2 - Metal strap installed under the power harness ground wire

Figure 2-16 Brush Block Installation

2.3 S/A HSDCD Signal Brush Block

CAUTION **DO NOT APPLY SIDEWAYS FORCE TO THE BRUSH BLOCK ASSEMBLIES. THE BRUSHES ARE BRITTLE AND WILL BREAK.**

1.) The Gantry contains three (3) brush blocks: two (2) for signal and one (1) for power.
2.) Install Signal Brush Block:
   a.) Hold the Brush Block in position, and align the brush tips with the slipring tracks.
   b.) Compress the brush tip springs against the slipring tracks, and hold the brush block against the brackets.
   c.) Install all four (4) screws, but do not tighten.
   d.) Adjust the brush blocks to center the top and bottom of the brush tips on the ring.
   e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)
   f.) Adjust the positioning set screws on the bracket to contact the brush block assembly. Set screws contain pre-loaded Loctite. Reference Figure 2-14 on page 30.
3.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.

2.4 Rotating Transceiver Board

NOTICE **Use proper ESD procedures when handling any circuit boards. Use the hardware from** Comment:, on page 20 **.**

1.) Install the rotating transceiver board on the inside surface of the slipring.

Note: The transceiver board has two coax cables connected to it.

2.) Install the five (5) 4-40 screws, and one (1) 1/4-20 ground screw, that fasten the transceiver PWB to the slipring.

CAUTION **Make sure the flat washers do not contact the ground plane. This will result in communication errors.**

3.) Connect the two (2) coax cables to J1 and J2 on the transceiver PWB.
4.) Connect the ground strap as shown in Figure 2-17.

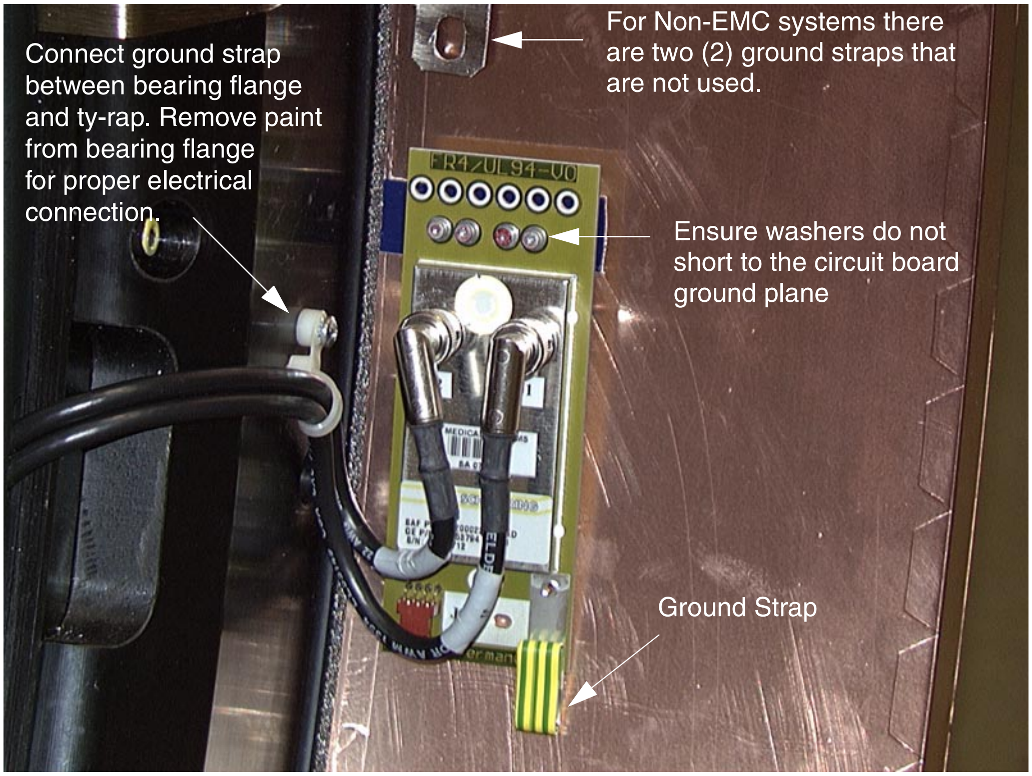


Figure 2-17 Rotating Transceiver Board. Non-EMC shown.

2.5 Rotating Terminator Board

NOTICE **Use proper ESD procedures when handling any circuit boards.**

1.) Locate the rotating terminator PWB on the inside surface of the slipring.

Note: The Terminator PWB has no cables attached to it.

2.) Install the two (2) 4-40 screws, and one (1) 1/4-20 ground screw, that fasten the terminator PWB to the slipring.

CAUTION **Make sure the flat washers do not contact the ground plane. This will result in communication errors.**

2.6 Stationary Transceiver Board

NOTICE **Use proper ESD procedures when handling any circuit boards.**

1.) Install the five (5) 4-40 screws that fasten the Stationary Transceiver PWB to the brush block assembly.

CAUTION **Make sure the flat washers do not contact the ground plane. This will result in communication errors.**

2.) Connect the two (2) coax cables to J1 and J2 on the transceiver PWB.

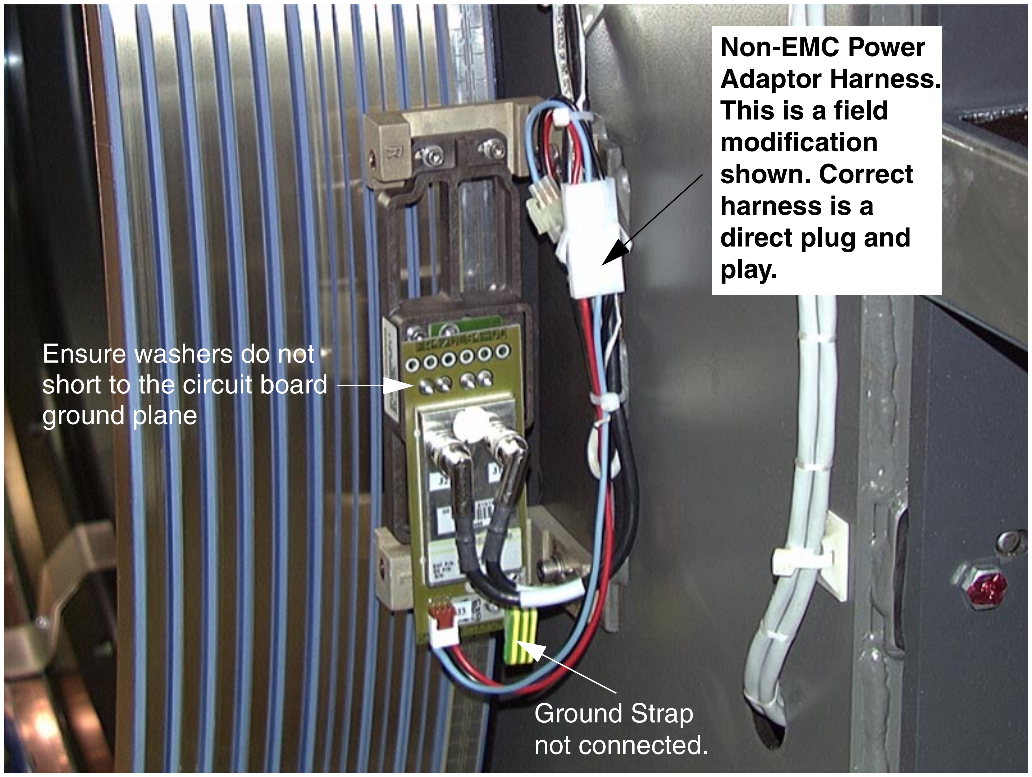


Figure 2-18 Stationary Transceiver Board. Non-EMC shown.

3.) Install Power harness adaptor for Non-EMC and very early HSA gantry's. Ty-Rap securely.

2.7 Stationary Terminator Board

NOTICE **Use proper ESD procedures when handling any circuit boards.**

Install the two (2) 4-40 screws that fasten the terminator PWB to the signal brush block assembly. Reference Figure 2-16 on page 31.

CAUTION **Make sure the flat washers do not contact the ground plane. This will result in communication errors.**

2.8 Rotating Transceiver Power Harness

2.8.1 46-XXXXXXGX Gantries

1.) Install rotating power harness.
   a.) Connect new power harness to OBC power supply. Reference Figure 2-19.
      1.) Pin 1 of transceiver to +15 vdc (+V2) of OBC power supply.
      2.) Pin 4 of transceiver to -15 vdc NOT USED.
      3.) Pins 2/3 of transceiver to common (-V2/+V3) of OBC power supply.

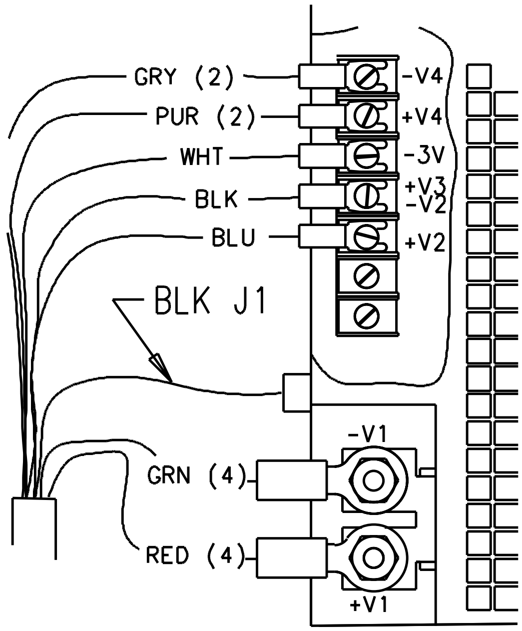


Figure 2-19 OBC Power Supply Power Harness Connections

Pin 1 of Transceiver is closest to the ground strap and/or center of the board.

   b.) Route power harness along bottom of OBC fan to rear of OBC chassis.
   Secure to existing harnesses with ty-raps.
   c.) Follow OBC backplane primary harness to casting opening.
      1.) Feed power harness through same opening in casting as the RCOM coax cables.
      2.) Secure to existing harnesses with ty-raps.
   d.) Route power harness through Rotating Harness cover slot following existing RCOM coax cabling. Reference Figure 2-20 on page 36.

2.) Proceed with steps 2.8.2 CTi and Lightspeed QXI Systems.

2.8.2 CTi and Lightspeed QXI Systems

1.) Remove ten (10) Phillips screws from the rotating harness cover. Reference Figure 2-20.
2.) Remove and route the rotating power harness along side the signal coax cables.
3.) Replace the ten (10) Phillips screws.
4.) Connect the Rotating power harness to the rotating transceiver board.

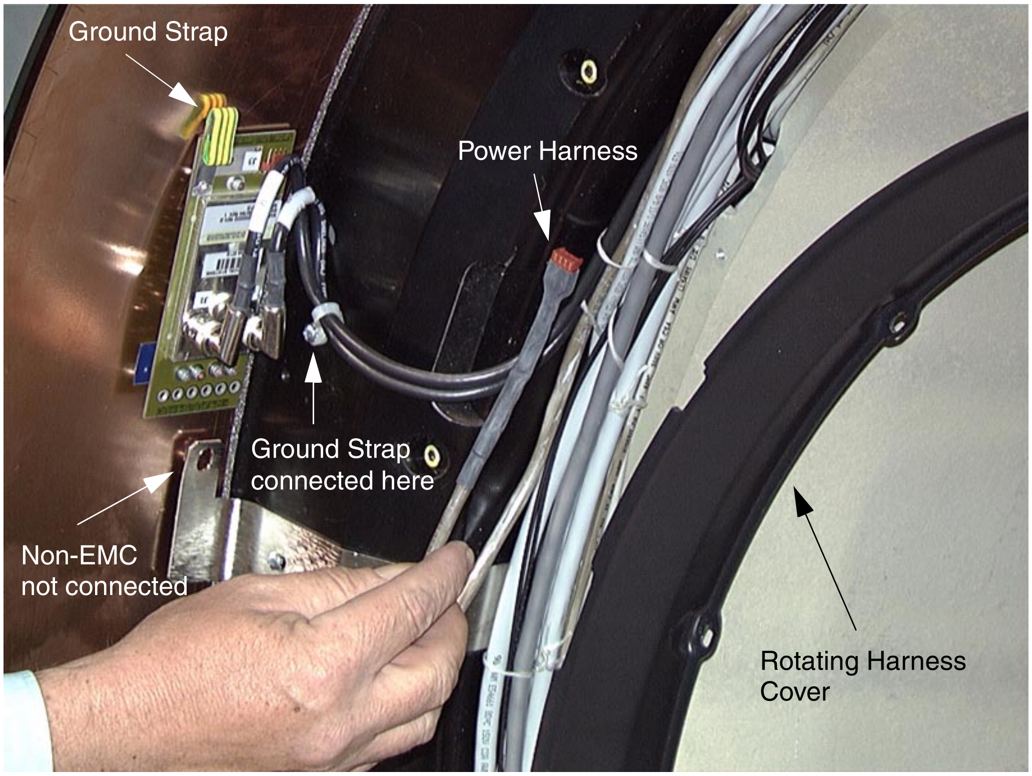


Figure 2-20 Rotating Transceiver Power Harness

2.9 S/A HSDCD (Lightspeed QXI ONLY)

2.9.1 S/A Stationary Power Pan Assembly

1.) Install new power pan assembly brackets.
2.) Connect fiber optic cables.
Use new fiber optic cable from transceiver to power pan, and route as shown in Figure 2-15 on page 30.
3.) Route coax and stationary transceiver cables along stationary bracket. Reference Figure 2-15 on page 30.

2.9.2 S/A HSDCD Slipring Antenna Replacement/Adjustment Procedure

NOTICE **HSDCD ANTENNA/RECEIVER AND TRANSMITTER ARE VERY ESD SENSITIVE. ALWAYS USE PROPER ESD PROCEDURES WHEN HANDLING THESE COMPONENTS, INCLUDING ADJUSTMENT PROCEDURES.**

This document describes the steps necessary to properly replace and adjust the position of the HSDCD antenna above the emitter traces on the rotating slipring. Accurate placement of the HSDCD antenna is important to achieve the optimal data transfer performance and to avoid contact between the ring and the stationary antenna. The optimal position of the HSDCD antenna above the emitter traces on the ring is 1.41 mm height (.060 inches), and centered. Under normal circumstances, the position of the HSDCD antenna should never need adjustment. Only adjust if there are indications of interference. Under no circumstances should the HSDCD antenna be allowed to contact the rotating components.

NOTICE **HSDCD ANTENNA/RECEIVER AND TRANSMITTER ARE VERY ESD SENSITIVE! ALWAYS USE PROPER ESD PROCEDURES WHEN HANDLING THESE COMPONENTS INCLUDING ADJUSTMENT PROCEDURES.**

1.) Attach the Antenna/Receiver assembly to the stationary bracket use proper ESD procedures. Reference Figure 2-15 on page 30.
2.) Loosen the axial adjustment screws and the radial adjustment screw for the HSDCD antenna.
3.) The HSDCD antenna should now be loose above the emitter. Insert the HSDCD adjustment tool between the HSDCD antenna and the emitter traces on the ring. The tool should fit snugly. The desired result is to space the HSDCD antenna above the emitter, at a height of 1.41 mm (.060 inches) above the emitter.

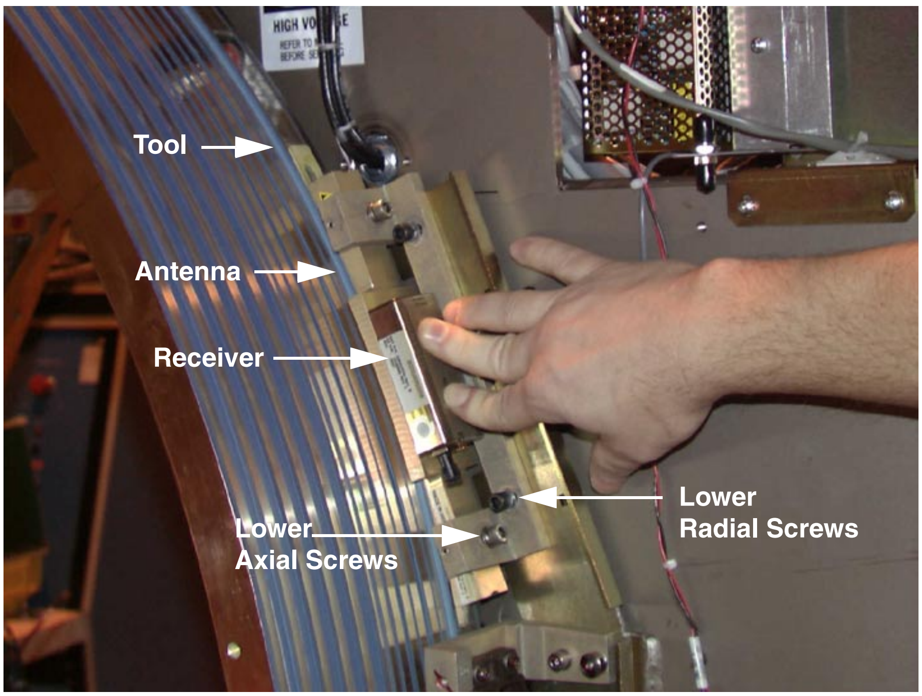


Figure 2-21 Setting HSDCD Antenna Gap

4.) Tighten the radial screws. Press lightly on the top of the HSDCD antenna as the adjustment screws are tightened.
5.) Carefully slide the adjustment tool from between the HSDCD antenna and the emitter.
6.) Using the alignment sights at each end of the HSDCD antenna, center it above the trace marks on the emitter PCB as shown.

Figure 2-22 HSDCD Antenna Centering

7.) While holding the position of the HSDCD antenna, tighten the axial alignment screws.

NOTICE Potential for Ring Damage — **Do not rotate the gantry with the adjustment tool installed. Damage to the delicate PCB traces will result. There is no method to repair ring boards in the field - a ring swap would be necessary!**

8.) Make sure the HSDCD has no power applied - there should be no LEDs on.

9.) Inspect the air-gap between the HSDCD antenna and the ring as the gantry rotates. It may be necessary to disable the gantry brake to rotate the gantry more easily. Look for clearances between the emitter and the HSDCD antenna. While rotating the ring, check that the emitter trace is aligned with the HSDCD. During rotation, no parts of the HSDCD antenna should contact the emitter surface.

Note: The stationary and rotating components must never touch, even with the gantry tilted.

- The run-out of the drum slipring traces should not exceed 0.83 mm axially, and 0.81 mm radially.
- Especially check clearances near the emitter solder and PCB connections.
- The HSDCD receiver has two LEDs. One LED indicates power is applied to the HSDCD and the other indicates the HSDCD is transmitting a signal.

10.) Re-install the slipring covers. Power-up the system. Replace the gantry rear cover and secure.

11.) Re-install the scan plane window. Verify proper operation by running verification scans.

12.) Run verification scans. Verification procedure should consist of:

- Observe diagnostic “DIP Stats” information. Before starting, log raw ring error count, FEC correctable count, and the date/time of the last file update.
- Run 5 stationary and 50 rotational scans with x-ray. The x-ray technique is not important. It is important to exit the exam, because this triggers the “DIP Stats” update.
- Observe diagnostic “DIP Stats” information. There should be no additional raw HSDCD ring errors or FEC correctable events.

2.9.3 S/A HSDCD Rotating Transmitter Power Measurements

1.) Locate the HSDCD Transmitter Power supply next to the transmitter on the inside of the rotating slip-ring assembly.

2.) Measure the DC voltage on the wires leading to the HSDCD Transmitter. The specification is +15 volts DC, +/- 2.0 volts. Use indicator LEDs to ensure proper operation.

2.9.4 S/A HSDCD Stationary Receiver Power Measurements

1.) Locate the HSDCD Receiver Power supply next to the Receiver near the top of the backside of the gantry frame for Non EMC systems. For EMC compliant systems check the STC power supply.

2.) Measure the DC voltage on the wires leading to the HSDCD Receiver. The specification is +12 volts DC, +/- 2.5 volts (+15 vdc +/- 0.25vdc for EMC systems). Use indicator LEDs to ensure proper operation.

Chapter 3
Test Procedures

Section 1.0
Slip-Ring Seasoning

1.) Select SERVICE DESKTOP Icon.
2.) Select DIAGNOSTICS.
3.) Select AXIAL CONTROL.
4.) Select AXIAL FUNCTIONAL TEST.
5.) Select CLOSED LOOP ROTATION.
6.) Select GANTRY SPEED **0.8** seconds.
7.) Select TEST DURATION **999**.

Note: It is important that you run 4,000 gantry rotations. This will season the ring and the brushes. Ignore any statistics that may be generated during this seasoning procedure.

Section 2.0
Testing Procedure

1.) Select SERVICE DESKTOP Icon.
2.) Select DIAGNOSTICS.
3.) Select DIAGNOSTIC DATA COLLECTION (DDC).
4.) Select ROTATING X-RAY OFF.
5.) Perform 100 scans each;
   - 4 second rotation speed.
   - 2 second rotation speed.
   - 1 second rotation speed.
   - 0.8 second rotation speed.

Verify no LAN watchdog time-outs, slipring view length errors were generated.

Section 3.0
Image Checks

Perform tube warm-up and fastcal.

Perform Image Series scans and verify Image Quality.

Chapter 4 Troubleshooting

Section 1.0 General Troubleshooting

GANTRY TYPE	ISSUE	ACTION
All	Cannot seat slip-ring drum on bearing flange.	Allow drum to achieve room temperature, 2 to 4 hours. This blue drum material has a high thermal coefficient. Ensure the EMI braid is not pinched between the drum and the bearing flange. A single strand will cause this issue.
All	Slip-ring Drum cracked during installation.	DO NOT USE MOUNTING BOLTS TO SEAT SLIPRING DRUM. Replace slip-ring drum.
EMC and Non EMC	Trouble aligning brushes.	Slip-ring drum not fully seated on bearing flange. Ensure no gap is present between the bearing flange and slip-ring drum.
EMC and Non EMC	Cannot install Stationary Transceiver ground screw. Board is mounted on an angle.	Ensure the HSDCD Receiver bracket is installed between the tilting frame and the brush block mounting brackets. This is the spacer.
All	No communications to the OBC.	Verify Rotation Power harness connected/ installed on rotating transceiver.
All	Elevated taxi link or black hole statistics.	Perform Slip-Ring Seasoning procedure.
All	Intermittent taxi link or black hole statistics.	Check all connections for tightness. Verify washers on transceiver connections are not shorting to ground plane of circuit board. Verify signal brushes are properly aligned. No “Blue Dust”. Verify Power Block Ground strap installed on ring #3.
Non EMC	Unable to scan at higher techniques. Above 100 ma	Incorrect Transceiver boards installed. Verify correct kit received, 2298034. Verify correct Transceiver installed, 2291808. Verify Power Block Ground strap installed on ring #3.
All	Fingerprints on slip-ring tracks.	Not an issue. Use alcohol to remove fingerprint during installation ONLY. Do this before the Slip-Ring Seasoning procedure.
LTSPD QXI	No DAS data collection possible.	Power not connected to TX or RX devices. Fiber optic cables not connected.
LTSPD QXI	DIP Taxi link errors	Receiver not properly aligned. Reference DIP Taxi Link T/S Section.

Section 2.0
LightSpeed QX/i DIP Taxi Link Errors

2.1 Symptoms

The symptoms are data collection errors resulting in scan aborts. In some cases scout scans work fine while axial scans fail. Others fail in all modes.

The following are the generated errors:

`Error 2300000064: TAXI data violation detected by forward error correction ERROR`

There is also a significant increase in DIP statistics.

Troubleshooting is difficult as there are also multiple symptoms.

- Stationary errors related to specific gantry position.
- Rotating errors only.
- Both stationary and rotating failures.
- Partial functionality with antenna/receiver purposely maladjusted, either skewed in the Z-axis or maximum gap adjustment.

2.2 Suggested Troubleshooting Process

1.) Review DIP Stats to establish baseline.

2.) Quantify failure modes. Stationary, rotating, positional. Use ScanDataPath diagnostics 50 passes or New Patient protocols.

3.) Scope power supplies, verify antenna alignment, fiber optic connections Power and Signal LEDs lit on both TX and RX units (LEDs always lit).

Note: Use extreme care with the fiber connections. These are keyed connectors with BNC style locking posts and can be easily damaged with excessive twisting force.

4.) Use slip-ring bypass procedure and verify no error generation. Do not rotate.

5.) Turn off gantry 120 VAC. Do not connect/disconnect power connector from TX or RX units with power applied (Power LED Lit). This can damage these units.

6.) Take resistance reading of HSDCD ring, on the ring itself, near the surface mount terminating resistors. Track to track ~ 18 ohms, same track opposite sides of terminator resistors ~ 2.5 ohms.

Note: If you read 9 ohms track to track then the terminator resistors are open or the track trace is open. If the track is open then you should read 48 ohms instead of 2.5 ohms. These failures require ring replacement.

7.) Carefully remove the transmitter. Do not touch the electronics as this part is ESD sensitive.

8.) Take resistance reading between the center pins of the 10 pin 2 row male connector located on the inside diameter of the ring. (What the transmitter mounts to).

Note: This resistance should be ~18 ohms, same as the track to track above. If it is significantly different then there is damage between this point and the HSDCD ring (emitter pcb). This would require ring replacement.

9.) Re-install transmitter, reconnect power connectors, turn on gantry 120vac.

10.) Start data collection and run multiple groups. 500 rotating passes without error is a high confidence result.

Note: If errors return after 10 minutes then the receiver may have a thermal breakdown. If test scans fail as before or immediately then proceed.

11.) Position HSDCD terminator resistors directly under the antenna and run stationary data collections. Manually rotate HSDCD terminator about 5 inch increments. Verify 10 passes per location.

Note: When a failing position is found you can misadjust the antenna by skewing, by lowering until it contacts the HSDCD ring, or by increasing the gap to maximum. If any of these provide improved performance it is most likely the Receiver has gain or noise immunity problems.

Chapter 5
Maintenance and Replacement Procedures

Section 1.0
Maintenance

After successful conversion kit installation you can expect slipring dust generation to begin. This is normal. In the first 250k rotations the dust generation will be the greatest. Simply use the same vacuum cleaner as used for the old style ETC slipring. There is no concern of old dust contamination.

- Simply vacuum the surface of the ring to remove dust deposits.
- Carefully remove the brush block assemblies, signal and power, and vacuum the dust deposits.
- Inspect the brush tips for wear. There is an arrow mark on each tip. When the tips wear to this mark replace the brushes.

CAUTION **Do not apply side forces to the brush tips. These tips are brittle and will crack requiring replacement.**

DO NOT

- Do not use alcohol or any chemicals to clean the ring or brush surfaces.
  - Alcohol will act as a binding agent allowing dust to accumulate in the crevices between rings.
  - Alcohol will remove the deposits, “Patina”, on the ring surfaces, which is the lubricant. This “Patina” is 1 to 3 mils thick and self renewing. This is normal.
  - Alcohol will soften the adhesive used to mount the HSDCD transmit ring. This will result in delamination of this PCB and require slip-ring replacement.
- Do not use “Craytex” or any other abrasive on the rings or brush tips.
  - Abrasives will damage the brass ring surface which is precision milled to be much smoother than glass.
  - Using abrasives on the brush tips will result in higher dust production as the tips will need to “re-seat” themselves to the ring surface.

CAUTION **Alcohol and Craytex should only be used in emergency service situations such as physical damage to the ring surface. Examples might be pitting of the brass ring surface due to high voltage arcing. Accidental nicks or scratches. Surface contamination due to high voltage oil or grease from hands and tools.**

Section 2.0
Replacement Procedures

2.1 S/A HSDCD Power Brush Block Module Replacement Details

1.) The Gantry contains three (3) brush blocks: two (2) for signal and one (1) for power.
   a.) Remove the grounding strap from the power brush block bracket before you remove the power brush block.
   b.) If you plan to replace the power brush block, remove the grounding strap from the power brush block and the power brush block bracket.
2.) Remove Brush Block:
   a.) Remove two (2) of the four (4) bolts, one on the top of the block and one on the bottom.
   b.) With one hand, hold the brush block against the brackets, and remove the remaining screws with the other hand.
   c.) Remove brush block from the brackets.
3.) Remove and install replacement modules:
   a.) Place brush block on a hard surface with the tips in the up position. There are five high power modules and three low power modules.
   b.) To remove a module, push the tab on the module out of the locking position. While pushing the tab out of the locking slot, pull the module up. Only pull the module out 2 mm, or just enough to keep the tab from snapping back into the slot. Repeat this procedure on the other side of the module.

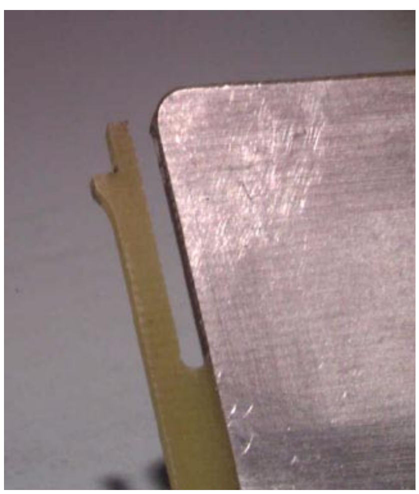


Figure 5-1 Power Module Locking Tab

Figure 5-2 Power Module Push Tab

c.) With the tabs on each side of the module out of the locking slot, pull the module straight out of the brush block housing.

Figure 5-3 Removing Power Module

d.) Align the replacement module in slots and push into brush block housing until the module tabs snap into the locking holes.

4.) Install Brush Block:

a.) Hold the Brush Block in position, and align the brush tips within the slip-ring tracks.

b.) Compress the brush tip springs against the slip-ring tracks, and hold the brush block against the brackets.

c.) Install all four (4) screws, but do not tighten.

d.) Adjust the brush blocks up against the set screws on the bracket to center the top and bottom of the brush tips on the ring.

e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)

5.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.

2.2 S/A HSDCD Signal Brush Block Module Replacement Details

1.) The Gantry contains three (3) brush blocks: two (2) for signal and one (1) for power.

Note: 2.) Remove the stationary Transceiver board to access and remove either of the signal brush blocks. Remove Brush Block:

a.) Remove two (2) of the four (4) bolts, one on the top of the block and one on the bottom.

b.) With one hand, hold the brush block against the brackets, and remove the remaining crews with the other hand.

c.) Remove brush block from the brackets.

3.) Remove and install replacement modules:

a.) Place the signal brush block on a hard surface with the tips down and remove the two screws that attach the module to the brush block body.

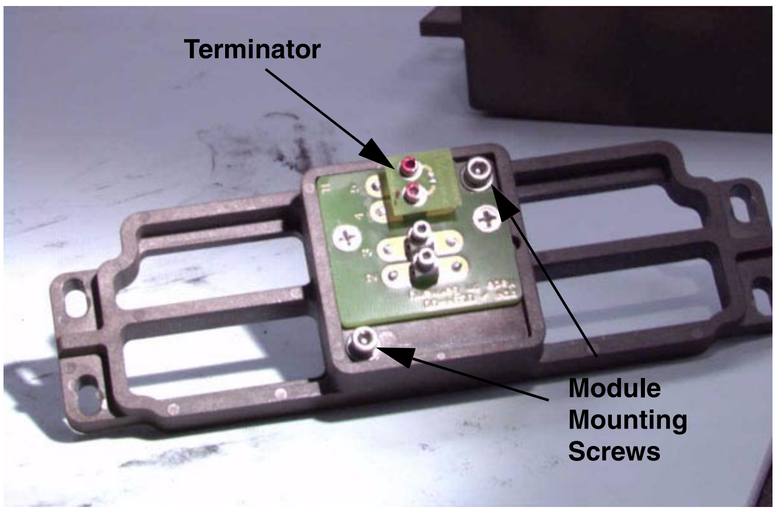


Figure 5-4 Signal Brush Block Module

b.) Insert the replacement module and tighten the screws.

c.) Make sure you install the terminator board as well.

4.) Install Brush Block:

a.) Hold the Brush Block in position, and align the brush tips with the slip-ring tracks.

b.) Compress the brush tip springs against the slip-ring tracks, and hold the brush block against the brackets.

c.) Install all four (4) screws, but do not tighten.

d.) Adjust the brush blocks up against the set screws on the bracket to center the top and bottom of the brush tips on the ring.

e.) Tighten all four (4) screws to 22 ±3in-lbs. (0.254 ±0.035 m-kg)

5.) Manually rotate Gantry while you check that the brush tips are centered between the track barriers.

2.3 S/A HSDCD Slip-Ring Receiver Replacement Procedure

DANGER USE AND FOLLOW TAG AND LOCK OUT PROCEDURES TO AVOID UN-COMMANDED GANTRY ROTATION. THERE ARE NO LETHAL VOLTAGES EXPOSED WITH THIS PROCEDURE.

1.) Remove the scan-plane window. Release and slide out rear gantry cover to allow access.
2.) Locate the HSDCD receiver on the rear gantry frame, upper left corner.
3.) Remove power connector first, then carefully remove the fiber optic connector.
4.) Remove the HSDCD receiver module by removing the two hex head screws.
5.) Replace the HSDCD receiver with the new component (receiver is ESD sensitive).
6.) Attach the fiber-optic cable and the power connector. The power connector is keyed for correct installation.
7.) Re-install the rear cover and apply power to the system.
8.) Verify proper operation by:
    a.) LEDs.
    b.) Observe the data indicator on the HSDCD receiver assembly is lit.
    c.) Run verification scans. Verification procedure should consist of:
        * Observe diagnostic “DIP Stats” information. Before starting, log raw ring error count, FEC correctable count, and the date/time of the last file update.
        * Run 5 stationary and 50 rotational scans with x-ray. The technique is not important. It is important to exit the exam, because this triggers the “DIP Stats” update.
        * Observe diagnostic “DIP Stats” information. There should be no additional raw HSDCD ring errors or FEC correctable events.

2.4 S/A HSDCD Slip-Ring Transmitter Replacement Procedure

Note: This item should already be installed on the slip-ring.

NOTICE HSDCD ANTENNA/RECEIVER AND TRANSMITTER ARE VERY ESD SENSITIVE! ALWAYS USE PROPER ESD PROCEDURES WHEN HANDLING THESE COMPONENTS INCLUDING ADJUSTMENT PROCEDURES.

1.) Remove the scan-plane window. Release and slide out rear gantry cover to allow access.
2.) Locate the HSDCD transmitter on the inside of the slip-ring.
3.) Remove power connector first, then carefully remove the fiber optic connector.
4.) Remove the HSDCD transmitter module by removing the two hex head screws.
5.) Replace the HSDCD transmitter with the new component (HSDCD transmitter is ESD sensitive).
6.) Attach the fiber-optic cable and the power connector. The power connector is keyed for correct installation.
7.) Re-install the rear cover and apply power to the system.
8.) Verify proper operation by:
    a.) Observe that the green LED indicator for power and yellow indicator for data is lit.
    b.) Run verification scans.
        * Observe diagnostic “DIP Stats” information. Before starting, log raw ring error count, FEC correctable count, and the date/time of the last file update.

* Run 5 stationary and 50 rotational scans with x-ray. The x-ray technique is not important. It is important to exit the exam, because this triggers the “DIP Stats” update.
* Observe diagnostic “DIP Stats” information. There should be no additional raw HSDCD ring errors or FEC correctable events.

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